

Education

King’s College London — Ph.D. Biomedical Engineering and Imaging Science	Oct 2023 – Oct 2027
Queen Mary University of London — M.Sc. Machine Learning (Distinction)	Sep 2022 – Sep 2023
SRM Institute of Science and Technology — B.Tech. Computer Science (Distinction)	Jul 2018 – May 2022

Representative Papers

Hsieh, I., S. S. Kundu, Vercauteren, T., & Dorent, R. (2027). SinkSLOT: Sinkhorn via Sparse Lifted Optimal Transport.	AAAI
Großkopf, E., S. S. Kundu et al. (2026). Redefining Instance Matching: A Unified Framework for Part-Aware Matching in Panoptic Segmentation Evaluation.	NeurIPS
S. S. Kundu et al. (2026). Instance Awareness of Multi-class Semantic Segmentation Loss Functions.	CV4CLINIC@CVPR
S. S. Kundu et al. (2025). Cluster Dice.	SPIE Medical Imaging
S. S. Kundu et al. (2024). Spinal osteophyte detection via robust patch extraction on minimally annotated x-rays.	IEEE ISBI
Naidu, R. & S. S. Kundu (2021). Improved variants of Score-CAM via Smoothing and Integrating.	RCV@CVPR
S. S. Kundu (2021). A Distributed Deep Learning Framework for Federated Big Medical Image Analysis.	IEEE Big Data

Relevant Industry Experience

Cosine (YC 23) — Machine Learning and Product Intern (Genie 3; Multilingual SWE-Bench)	Summer 2025
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Representative Software & Open Source Contributions

Panoptica — [Maintainer] Instance-wise segmentation metrics	33★
Ultraflwr — [Author] Federated learning on the edge	30★
torchsparsegradutils — [Maintainer] Utilities for PyTorch sparse tensors	40★
MONAI — [Core contributor] to the Medical Open Network for AI (MONAI)	8.5k★

Representative Fellowships & Grants

Google Summer of Code (GSoC) — Neuroinformatics Unit (1,141 of 23,371 proposals selected, ~4.9%)	2026
Claude for Open Source — Anthropic (open source maintainer grant, \$1,200)	2026
Codex for Open Source — OpenAI (open source maintainer grant, \$1,200)	2026
MRC DTP Postgraduate Studentship — King’s College London (<2% selection, ~£205k)	2023
BDI Summer Internship — University of Oxford (1 of 4 selected)	2023
Summer Research Internship — IIT Gandhinagar (<0.008% selection)	2023
MITACS Globalink Research Internship — TRIUMF (UVic) (<3% selection)	2021

Representative Awards

SPIE Medical Imaging Travel Grant Award (\$1000)	2025
IEEE ISBI’24 Best Student Poster Award Finalist (Top 8 of 717)	2024
UKRI Fast Start: Innovation Grant — Co-Applicant (~£50k)	2022

Programming & Software Development

Languages: Python, C++, Bash, L^AT_EX
ML: PyTorch, JAX, NumPy, CUDA
Tools: Git, Slurm, Docker, Kubernetes, macOS, Linux

Graduate Teaching

Image and Signal Processing (Prof. James Housden)	Spring 2026
Python (Dr. Marc Modat and Dr. Jonathan Shapey)	Spring 2026
Journal Club — Biomedical Engineering (Prof. Monica Agromayor)	Winter 2024 & Spring 2025
Statistics — Biomedical Engineering (Prof. James Housden)	Spring 2025

Supervision

Yingfang Tao — KCL BSc Thesis Supervision	2025–26
Jai Pradeep Awasthi and Pranav Rustagi (LNMIIT) with Yang Li	Summer 2025
In2Stem (4 High School students) with Lorena Garcia-Foncillas Macias	Summers 2024/25/26

Invited Talks

Adapting cellfinder to single-channel and 2.5-D data (NIU Open Software Summer School, SWC London, UCL.)	2026
Exploring Instance Imbalance in Medical Semantic Segmentation (Voxel51 Visual AI in Healthcare Meetup)	2024

Academic Service

Programme Committee: BrainWorks @ MICCAI 2026
Reviewing: MICCAI, ICML, NeurIPS, ICLR, AISTATS, IEEE-ISBI.